INCH-POUND

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MILITARY SPECIFICATION

HOSE ASSEMBLY, ELASTOMERIC:

FUEL DISCHARGE, COLLAPSIBLE

This specification is approved for use by the US Army Belvoir Research, Development, and Engineering Center based upon currently available technical information but it has not been approved for promulgation as a coordinated revision of MIL-H-53073. It is subject to modification. However, pending its promulgation as a coordinated military specification, it may be used in acquisition.

1. SCOPE

1.1 Scope. This specification covers collapsible, elastomeric discharge hose assemblies.

1.1 Classification. The collapsible discharge hose assemblies are of the following sizes and types as specified:

Size (Inside Diameter)

2-inch
4-inch
6-inch

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA, 22060-5606 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4720

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Types

0 - Without couplings
A - Cam-Locking type coupling
B - Single groove, snap lock couplings
C - Single groove, bolted couplings

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

QQ-S-766	- Steel Plates, Sheets, and Strip-Corrosion Resisting.
VV-F-800	- Fuel Oil, Diesel.
WW-C-440	- Clamps, Hose, (Low Pressure).

MILITARY

MIL-H-775	- Hose, Hose Assemblies; Rubber, Plastic, Fabric, or Metal (Including Tubing); Associated Hardware Packaging Of.
MIL-G-3056	- Gasoline, Automotive, Combat.
MIL-T-5624	- Turbine Fuel, Aviation, Grades JP-4 and JP-5.
MIL-C-10387	- Couplings, Clamp, Pipe, with Bolts and Synthetic Rubber Gasket, for Pipe and Tube.
MIL-F-46162	- Fuel, Diesel, Referee Grade.
MIL-T-83133	- Turbine Fuel, Aviation, Kerosene Type, Grade JP-8.

STANDARDS

FEDERAL

FED-STD-595	- Colors Used in Government Procurement.

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-889	- Dissimilar Metals.

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MIL-STD-1472 - Human Engineering Design Criteria for Military Systems, Equipment and Facilities.
MS27021 - Coupling Half, Quick Disconnect, Cam-Locking Type, Male, Hose Shank, Type II.
MS27025 - Coupling Half, Quick Disconnect, Cam-Locking Type, Female, Hose Shank, Type VI.
MS27028 - Coupling Half, Quick Disconnect, Cam-Locking Type, Cap, Dust, Type IX.
MS27029 - Coupling Half, Quick Disconnect, Cam-Locking Type, Plug Dust, Type X.
MS27030 - Gasket, Coupling Half, Quick Disconnect, Cam and Locking Type.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 380 - Rubber Hose Testing.
D 381 - Existent Gum in Fuels by Jet Evaporation.
D 412 - Rubber Properties in Tension.
D 413 - Rubber Property - Adhesion to Flexible Substrate.
D 439 - Automotive Gasoline.
D 471 - Rubber Property - Effect of Liquids
D 518 - Rubber Deterioration - Surface Cracking.
D 910 - Aviation Gasolines.
D 1053 - Measuring Rubber Property - Stiffening at Low Temperature Using a Torsional Wire Apparatus.
D 1149 - Rubber Deterioration - Surface Ozone Cracking in a Chamber (Flat Specimens).
D 3389 - Coated Fabrics Abrasion Resistance (Rotary Platform, Double-Head Abrader).

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia PA 19103.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

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2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, (except for related associated detail specifications, specification sheets or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. Unless otherwise specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.2 Material. Material shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification.

3.2.1 Material deterioration prevention and control. The hose assemblies shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operating and storage environments to which the hose assemblies may be exposed.

3.2.1.2 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.2.2 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with component and subcomponent, and shall make information available upon request to the contracting officer or designated representative.

3.2.3 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the hose assemblies may be newly fabricated from recovered materials to the maximum extent practicable, provided the hose assemblies produced meets all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the hose assemblies.

3.3 Design and construction. The hose assemblies shall consist of collapsible, reinforced synthetic rubber hose with attached cam-locking type coupling halves, or grooved end fittings.

3.3.1 Hose. The hose shall be constructed of a compounded inner tube, a synthetic fiber reinforcement, and a compounded cover.

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3.3.2 Tube. The tube shall be fabricated from a synthetic compound. The compound shall be resistant to aromatic hydrocarbon fuels, and shall not have any deleterious effects upon all types and grades of fuels conforming to MIL-G-3056 and ASTM D 439 (automotive gasoline), ASTM D 910 (aviation gasoline), MIL-T-5624 and MIL-T-83133 (jet fuels), and VV-F-800 and MIL-F-46162 (diesel fuel). The tube shall be smooth, free from pits, and of uniform thickness. The thickness of the tube shall be not less than 1/16-inch.

3.3.3 Cover. The cover shall be fabricated from a synthetic compound. The cover shall be uniform in thickness and shall be not less than 1/16-inch. Unless otherwise specified (see 6.2), the color of the cover shall be sand matte in accordance with FED-STD-595, color chip 33446.

3.3.4 Coupling halves. The coupling halves shall be type A, B, or C, as specified (see 1.2 and 6.2). Coupling halves shall be securely attached to the hose to form a permanent electric bond with the static wires. Type 0 hose assembly is designated as without couplings.

3.3.4.1 Type A, cam-locking types. Type A cam-locking type coupling halves with specified gaskets shall be furnished with each size of hose as shown in table I, except that the working pressure rating shall be 150 psi (pounds per square inch). Dust plugs, caps and gaskets shall be in accordance with MS27028, MS27029, and MS27030, respectively.

3.3.4.2 Type B and C, single grooved couplings.

3.3.4.2.1 Hose to grooved adapter. For type B and C couplings, a one piece aluminum adapter shall be provided with grooved end conforming to MIL-C-10387, figure 1. For each size, the dimensions shown in MIL-C-10387 for the equipment, nominal pipe size shall be used. The nipple shall conform to the dimensions shown on MS27021, for the "C" length only.

3.3.4.2.2 Grooved coupling. The grooved couplings shall conform to MIL-C-10387, class A. For each size, the equivalent nominal pipe size from MIL-C-10387 shall be used.

3.3.5 Hose clamps. Each cam-locking type coupling half and single grooved coupling shall be banded to the hose with clamps conforming to WW-C-440, type H, except the clamp material shall be austenitic stainless steel conforming to QQ-S-766, class 201, condition A, and the hydraulic test pressure shall be 225 psi. Quantity and size of clamps for each size hose shall be as shown in table I. When installed and locked, the clamps shall have no protruding ends.

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TABLE I. Coupling halves and clamps.

Hose Size Nominal	Couplings		Clamps	
	Female	Male	Size width	Quantity per coupling
2	MS27025-11	MS27021-11	5/8	2
4	MS27025-17	MS27021-17	3/4	2
6	MS27025-21	MS27021-21	3/4	3

3.3.6 Reinforcement. The reinforcement material shall be one or more plies of braided or spiraled synthetic fiber cord or fabric which shall be evenly and firmly fabricated and thoroughly impregnated with synthetic rubber compound which shall cause the plies to adhere firmly to each other and to the tube and cover. The reinforcement material shall be applied by braiding, looming, or spiral winding twist to provide a balanced design. The reinforcement shall be evenly and firmly fabricated, and shall be free from defects, dirt, knots, lumps, and irregularities of twist so that the completed hose shall comply with all the applicable requirements herein.

3.3.7 Static wires. Two helically wound, static electric bonding wires shall be embedded within or between plies to form an angle of not less than 45 degrees to the longitudinal axis of the hose. The wires shall not break when the hose is coiled or subjected to internal or external pressure. The wires may be either braided or stranded and shall have an area equivalent to:

Stranded - 7 strands no. 31B and S gauge (0.0089-inch diameter)

Braided - 48 strands no. 36B and S gauge (0.005-inch diameter)

When tested as specified in 4.5.2.1, the static wire shall be continuous to effect an electric bond between ends of the hose (maximum resistance not to exceed 1.5 ohms per linear foot of hose).

3.4 Physical and chemical properties.

3.4.1 Hydrostatic pressure. When tested as specified in 4.5.2.1, the hose with couplings attached shall not leak nor show any imperfections in either the hose or couplings.

3.4.1.1 Length change and twist. When tested as specified in 4.5.2.8, the length of hose shall not change more than ±3 percent and shall not twist either clockwise or counterclockwise more than 1 degree per foot of length.

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3.4.1.2 Minimum burst. When tested as specified in 4.5.2.2, each type of hose shall have a minimum burst pressure as specified in table II.

TABLE II. Burst and working pressure.

Size I.D. (inside diameter) (inches nominal)	Burst pressure psi (minimum)	Working pressure psi
2	400	100
4	600	150
6	600	150

3.4.2 Tensile strength. When tested as specified in 4.5.2.3.1, the tensile strength of the tube shall be not less than 1,250 pounds force per square inch psi and the tensile strength of the cover shall be not less than 1,000 psi. When tested as specified in 4.5.2.3.2, the tensile strength of the tube and cover after fuel immersion shall be not less than 40 percent or less than 600 psi (whichever is greater) of the tensile strength obtained before immersion.

3.4.2.1 Tensile strength after water immersion. When tested as specified in 4.3.3, the tensile strength of the tube and cover after 14-day water immersion shall be not less than 40 percent or less than 600 psi (whichever is greater) of the tensile strength obtained before immersion. The tensile strength after 42-day water immersion shall be not less than 25 percent or less than 300 psi (whichever is greater) of the tensile strength before immersion.

3.4.3 Ultimate elongation. The ultimate elongation of the tube and cover shall be not less than 200 percent. When tested as specified in 4.5.2.3.2, the ultimate elongation of the tube and cover shall be not less than 40 percent of the original elongation obtained before fuel immersion.

3.4.3.1 Ultimate elongation after water immersion. When tested as specified in 4.3.3, the ultimate elongation after 14-day water immersion, shall be not less than 40 percent of the original elongation obtained before immersion. The ultimate elongation, after 42 days, shall be not less than 25 percent of the original elongation obtained before immersion.

3.4.4 Adhesion.

3.4.4.1 Original. When tested as specified in 4.5.2.3.3, the original adhesion between the tube and plies, and between the cover and the plies, and between the cover and end reinforcement, shall be not less than 10 pounds force (lb_f).

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3.4.4.2 After fill test. When tested as specified in 4.5.2.3.4, the adhesion between the tube and plies, between the plies, and between the cover and the plies and between the cover and end reinforcement, shall be not less than 8 lb_f .

3.4.5 Volume increase. When tested as specified in 4.5.2.4, the volume increase of the tube specimen after immersion in test fluid shall not exceed 40 percent, and the volume increase of the cover specimen shall not exceed 70 percent.

3.4.6 Low temperature flexibility. When tested as specified in 4.5.2.5, the relative torsional modules for the inner tube and the outer cover shall not be greater than 5 at -32 °C (-25.6 °F).

3.4.7 Existent gum content. When tested as specified in 4.5.2.6, the remainder shall be not more than 20 milligrams per 100 milliliters of test fluid.

3.4.8 Ozone resistance. When tested as specified in 4.5.2.7, the hose cover material shall show no evidence of cracking, checking, or irregularities at the end of the exposure time when viewed under 7 power magnification.

3.5 Quantities, dimensions, and weights. The quantities and length of the hose shall be as specified (see 6.2) and shall be the actual hose length exclusive of the couplings. A tolerance in length of ±2 percent shall be permitted. This hose shall be furnished in the diameters and weights shown in table III.

TABLE III. Diameters and weights,

Size (I.D.) (inches +1/32-inch)	Outside diameter (inches at ends)	Pounds per foot (maximum)
2	2-9/16, +1/8 or -1/16	1-1/4
4	4-3/4, +1/8 or -1/16	1-3/4
6	6-3/8, +1/8 or -1/16	2-5/8

3.6 Identification marking. Each length of hose shall be branded with:

Manufacturer's name or trademark
Quarter and year of manufacture
Manufacturer's serial number
Type
Working pressure
Hose, Liquid Petroleum Fuels, Dispensing, Collapsible

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The hoses shall be branded with the manufacturer's standard working pressures for the specific type and size hose supplied, and shall be rated at the working pressures indicated in table II. The letters shall be molded, either recessed or in relief, shall be of a contrasting color to the hose color and shall be not less than 1/4-inch high. Letter characteristics shall be in accordance with MIL-STD-1472 (5.5.1). Hoses shall be branded at 10-foot intervals or by continuous-impression branding, provided that the pattern shall be repeated every 5 feet or less.

3.7 Abrasion resistance. When tested as specified in 4.5.2.10, the hose sample shall show no signs of the textile reinforcement or abrading through the exterior coating after 100,000 cycles using H-22 abrasion wheels.

3.8 Workmanship. The hose assemblies shall conform to the quality requirements specified herein with no cuts, nicks, bruised, blisters, sharp edges, or lumps, and shall be clean and smooth. Each hose assembly shall be free from any other workmanship deficiencies that could impair the function or serviceability of the hose or end fitting in its intended use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the document where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this document shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the document shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

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4.1.2 Component and material inspection. The contractor is responsible for ensuring that components and materials are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection of packaging (see 4.6).

4.3 First article inspection.

4.3.1 Examination. The first article hose assembly shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The first article hose assembly shall be tested as specified in 4.5.2.1 through 4.5.2.10 inclusive. Failure of any test shall be cause for rejection of the hose assembly.

4.3.3 Tensile strength and ultimate elongation after water immersion. Tensile strength and ultimate elongation after water immersion shall be determined in accordance with ASTM D 471. The immersion test shall be performed in accordance with ASTM D 471, utilizing water as a test fluid. One specimen shall be subjected to water immersion test for 14 days at 23 °C. Nonconformance to 3.4.2.1 and 3.4.3.1 shall constitute failure of this test.

4.4 Quality conformance inspection.

4.4.1 Lot. Each lot shall consist of all units of hose assemblies of the same size manufactured under the same conditions and at the same time, and offered to the Government at one time.

4.4.2 Sampling. Sampling of hose and hose assemblies for examination and tests shall be in accordance with MIL-STD-105, table I and table IIa. A lot shall be accepted when 0 defects are found and rejected when 1 or more defects are found.

4.4.3 Examination. Samples selected in accordance with 4.4.2 shall be examined as specified in 4.5.1.

4.4.4 Tests.

4.4.4.1 Individual. Each hose assembly shall be tested as specified in 4.5.2.1. Failure of the test shall be cause for rejection.

4.4.4.2 Samples. Samples selected in accordance 4.4.2 shall be tested as specified in 4.5.2.2 through 4.5.2.10, inclusive.

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4.5 Inspection procedure.

4.5.1 Examination. The hose assembly shall be examined as specified herein for the following defects:

101. Materials not as specified (see 3.2).
102. Materials not resistant to corrosion and deterioration, or treated to be resistant to corrosion and deterioration of the applicable storage and operating environments (see 3.2.1).
103. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other (see 3.2.1.1).
104. Contractor does not have documentation available for identification of materials, material finishes, or treatment (see 3.2.2).
105. Used, rebuilt or remanufactured components, pieces, or parts incorporated in the hose assembly (see 3.2.3).
106. Design and construction not as specified (see 3.3).
107. Color of cover not as specified (see 3.3.3).
108. Hose clamp not as specified (see 3.3.4).
109. Couplings not as specified (see 3.4).
110. Diameters and weights not as specified (see 3.5).
111. Identification marking missing, illegible, or not as specified (see 3.6).
112. Workmanship not as specified (see 3.8).

4.5.2 Tests.

4.5.2.1 Hydrostatic. Each length of hose, with fittings such as couplings and adapters attached, shall be subjected to hydrostatic pressure test in accordance with ASTM D 380 and shall be 225 psi for all sizes. The electric bond shall be determined while full test pressure is applied. Nonconformance to 3.3.7 and 3.4.1 shall constitute failure of this test. Testing may be performed without specified couplings and shall be not more than 1.5 ohms per linear foot of hose.

4.5.2.2 Minimum burst. A sample of not less than 3 feet nor more than 6 feet shall be cut from the one length selected from each 100 lengths for burst and physical test. The hose shall be subjected to a hydrostatic test pressure in accordance with ASTM D 380 and as specified in table II. The pressure shall be supplied at a uniform rate sufficient to develop the test pressure in 10 seconds, ±2 seconds. Nonconformance to 3.4.1.2 shall constitute failure of this test. The remaining section of hose from which the test sample has been taken shall be accepted as a full length, provided the sample meets all the requirements of this specification. Testing may be performed without specified couplings.

4.5.2.3 Tensile strength, elongation, and adhesion.

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4.5.2.3.1 Tensile and elongation specimens. Tensile and elongation specimens shall be tested in accordance with ASTM D 412 method A. Die "C" shall be used for cutting test specimens. The tube specimens shall be buffed to not less than 0.050-inch thickness. Nonconformance to 3.4.2 and 3.4.3 shall constitute failure of this test.

4.5.2.3.2 Tensile strength and ultimate elongation after fuel immersion. Tensile strength and ultimate elongation after fuel immersion shall be determined in accordance with ASTM D 471. Nonconformance to 3.4.2 and 3.4.3 shall constitute failure of this test.

4.5.2.3.3 Original adhesion test. A dry specimens of hose material shall be tested in accordance with ASTM D 413 machine method. Nonconformance to 3.4.4.1 shall constitute failure of this test.

4.5.2.3.4 Adhesion after filling. A 12-inch length of the hose shall be stoppered and filled with the test fluid specified in 4.5.2.3.1. The section of hose with the test fluid shall be maintained at a temperature of 23, ±3 °C (160, ±2 °F) for not less than 48 hours. After cooling to 73, ±5 °F, ring specimens shall be cut from the 12-inch length of hose and tested in accordance with ASTM D 413 machine method within 15 minutes after the removal of the test fluid. Nonconformance to 3.4.4.2 shall constitute failure of this test.

4.5.2.4 Volume increase. Volume increase test shall be determined in accordance with ASTM D 471 with the test fluid specified in 4.5.2.3.2 at 23 °C for 46 hours. Nonconformance to 3.4.5 shall constitute failure of this test.

4.5.2.5 Low temperature flexibility. The test for low temperature flexibility shall be made in accordance with ASTM D 1053. The low temperature test shall be determined after the samples have been conditioned 7 days at -32 °C (-25.6 °F). Nonconformance to 3.4.6 shall constitute failure of this test.

4.5.2.6 Existent gum. A test sample of hose, not less than 14 inches long, shall be selected and the bottom plugged with a clean corrosion-resisting steel cylinder 2 inches long secured in place with a clamp. The sample shall be filled to within 2 inches of the top with test fluid specified in 4.5.2.3.1. The top end of the hose shall be plugged in a manner similar to the bottom. The sample shall then be stored in a vertical position for 7 days at an ambient temperature of 38, ±1 °C (100, ±2 °F). At the end of each 24 hours, the fuel in the sample shall be agitated for 5 minutes by moving the hose back and forth from the vertical to a horizontal position at a rate of 2 cycles per minute. At the end of the 7-day storage period, the fuel shall again be agitated in the sample for 5 minutes and immediately removed from the hose. The removed fuel

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shall be tested for existent gum in accordance with the air-jet solvent wash method of ASTM D 381. A blank shall be run on the test fluid at the same time and by the same method. The existent gum of the blank fluid shall be subtracted from the existent gum obtained from the test fluid removed from the hose. Nonconformance to 3.4.7 shall constitute failure of this test.

4.5.2.7 Ozone resistance. Specimens of the cover, prepared as described in ASTM D 518, procedure A, shall be mounted in a 20 percent elongated position and tested in accordance with ASTM D 1149. After conditioning for 24 hours in a ozone-free atmosphere, the mounted specimens shall be exposed for 336 hours at 40, ±1 °C (104, ±1 °F), at an atmosphere containing 50 parts, ±10 parts, per hundred million (pphm) of ozone. Upon completion of the test, the specimens shall be examined. Nonconformance to 3.4.8 shall constitute failure of this test.

4.5.2.8 Hydrostatic pressure with kerosene. A sample of hose, not less than 3 feet and not greater than 6 feet with fittings such as couplings and adapters attached, shall be subjected to a hydrostatic pressure test in accordance with ASTM D 380 and as specified in 4.5.2.1, except that kerosene shall be used as the test fluid. Nonconformance to 3.4.1 shall constitute failure of this test.

4.5.2.9 Length change and twist. After a pressure of 10 psi is applied and maintained, a measurement of the original length of each length of hose shall be made, with the hose laid out in a straight horizontal position. With a crayon or soft pencil, make a mark on the top surface of the hose at each end adjacent to the coupling. These marks are used as a means of measuring the amount of twist during the test. The pressure shall then be increased to 100 psi and held for not less than 30 seconds. With the pressure still maintained, the final length measurement and the final twist measurement shall be taken. The percent change in length shall be calculated as plus for increase and minus for decrease in length. Nonconformance to 3.4.1.1 shall constitute failure of this test.

4.5.2.10 Abrasion resistance. Three test samples shall be cut from the hose with the abrasion testing performed on a cover side. The abrasion test shall be performed according to ASTM D 3389, method A, using wheel, H-22. Nonconformance to 3.8 shall constitute failure of this test.

4.6 Inspection of packaging. The packaging shall be examined to determine compliance with the quality assurance provisions of MIL-H-775 and with the requirements of section 5 herein.

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5. PACKAGING

5.1 Preservation. Preservatives shall not be used for hoses covered by this specification. Preservation shall be level A or level C as specified (see 6.2)., and shall be in accordance with the applicable preservation requirements of MIL-H-775.

5.2 Packaging. Packaging shall be level A, level B, or level C as specified (see 6.2), and shall be in accordance with the applicable packaging requirements of MIL-H-775.

5.3 Marking. Marking shall be in accordance with all the marking requirements of MIL-H-775.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The hose assemblies covered by this specification are intended for transferring automotive and aviation gasoline, JP-4 or JP-5 jet fuel, or diesel fuels.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of the specification.
b. Size required (see 1.2).
c. Type of coupling (see 1.2).
d. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
e. When a first article is required for inspection and approved (see 3.2 and 4.3).
f. Color of cover, if other than as specified (see 3.3.3).
g. Quantity and length of hose assembly (see 3.5).
h. Degree of preservation and packaging required (see 5.1 and 5.2).

6.3 First article. When a first article inspection is required, the items should be a preproduction model. The first article should consist of one hose assembly. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of the first article test results and disposition of the first articles. Invitation for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders

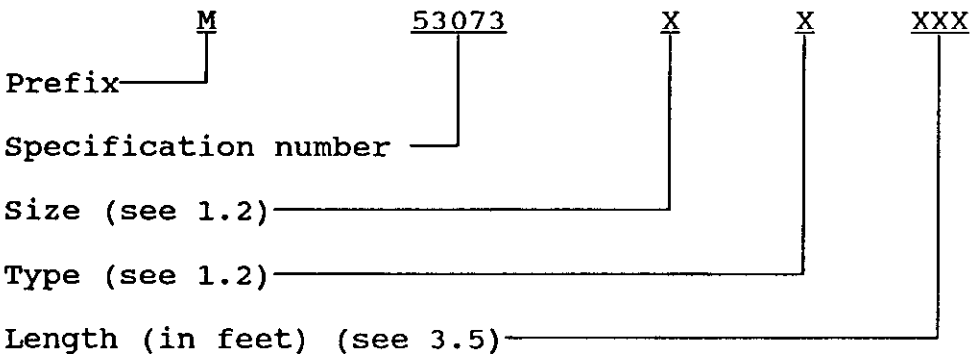
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offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Subject term (key word) listing.

Collapsible
Discharge
Fuel
Hose

6.5 Part or identifying number (PIN). The PIN to be used for hose assembly acquired to this document are created as follows:



Example - M530732A500 (A hose assembly conforming to MIL-H-53073 having a 2-inch inside diameter with cam-locking type couplings and a length of 500 feet.)

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - ME

Preparing activity:
Army - ME

Project 4720-A042

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-H-0053073A(ME)	2. DOCUMENT DATE (YYMMDD) 920514

3. DOCUMENT TITLE
Hose Assembly, Elastomeric: Fuel Discharge, Collapsible

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (if applicable) (2) AUTOVON	7. DATE SUBMITTED

8. PREPARING ACTIVITY

a. NAME Carolyn B. Johnson	b. TELEPHONE (Include Area Code) (1) Commercial (703) 704-3468	(2) AUTOVON 654-3468
c. ADDRESS (Include Zip Code US Army Belvoir RDE Center ATTN: STRBE-TSE Fort Belvoir, VA 22060-5606	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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